

# A component flux gives a determinant difference

MathlibAnnex.Piola.divergence\_componentFlux\_eq\_det\_sub

## theorem

Expresses the change from one output-coordinate perturbation as a divergence.

## Statement

Let  $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be  $C^2$ , let  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$  be  $C^1$ , and fix  $i \in \{1, \dots, n\}$ . Let  $e_j$  be the  $j$ th standard coordinate vector and let  $A$  denote an arbitrary real  $n \times n$  matrix. Write  $g_k(x) = (g(x))_k$  and

$$C_i(A)_j = \det(A[i \leftarrow e_j^T]).$$

Define the perturbed map and its flux by

$$\begin{aligned} g^{(i)}(x) &= g(x) + \varphi(x)e_i, \\ V(x) &= \varphi(x)C_i(Dg(x)), \quad V_j(x) := (V(x))_j = \varphi(x)C_i(Dg(x))_j. \end{aligned}$$

Then, for every  $x$ ,

$$\operatorname{div} V(x) := \sum_{j=1}^n \partial_j V_j(x) = \det Dg^{(i)}(x) - \det Dg(x).$$

## Assumptions

The differentiability assumptions are exactly  $C^2$  on  $g$  and  $C^1$  on  $\varphi$ . The row  $i$  and point  $x$  are arbitrary. Compact support and invertibility are not required for this pointwise identity.

## Conclusion

The determinant increment is the divergence of the explicitly defined vector field  $V$ . If  $\varphi$  has compact topological support, then so does  $V$ : with  $\operatorname{supp} w := \overline{\{x : w(x) \neq 0\}}$ , one has  $\operatorname{supp} V \subseteq \operatorname{supp} \varphi$ . This separate support consequence is what permits the subsequent integration.

## Notes

Only the  $i$ th output coordinate changes, so only row  $i$  of the derivative matrix changes. The row convention for the cofactor is essential to the identity.

## Proof route

Apply the product rule, cancel the cofactor divergence by Piola, and recognize the remaining sum as a row-replacement determinant.

## ▼ Proof steps

1. Put  $A = Dg(x)$  and let  $d\varphi$  be the row with entries  $\partial_j\varphi(x)$ . Coordinate vectors satisfy  $v = \sum_j v_j e_j$ , and determinant multilinearity in one row consequently gives

$$\det(A[i \leftarrow v]) = \sum_j v_j C_i(A)_j.$$

The finite-sum row-expansion lemmas justify this identity without an invertibility assumption.

2. The cofactor field is  $C^1$ : it is a polynomial in the entries of  $Dg$ , which is  $C^1$  because  $g$  is  $C^2$ . The product derivative is therefore valid, and Piola supplies its zero term:

$$\begin{aligned} \operatorname{div} V(x) &= \sum_j \partial_j \varphi(x) C_i(A)_j + \varphi(x) \sum_j \partial_j (C_i(Dg))_j(x) \\ &= \sum_j \partial_j \varphi(x) C_i(A)_j = \det(A[i \leftarrow d\varphi]). \end{aligned}$$

3. Differentiation gives  $Dg^{(i)}(x) = A + e_i d\varphi$ , whose row  $i$  is  $A_i + d\varphi$  and whose other rows agree with  $A$ . By linearity in that row,

$$\det Dg^{(i)}(x) = \det A + \det(A[i \leftarrow d\varphi]).$$

Subtract  $\det A$  and insert the previous expression for the divergence. For the later support assertion,  $\varphi(x) = 0$  implies  $V(x) = 0$ , so the support of  $V$  is contained in that of  $\varphi$ ; taking closures preserves the inclusion.

## Main citations

- Exact declaration and its source — `MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`
- Divergence-free cofactor field — `MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero`
- Cofactor row by replacement — `MathlibAnnex.Piola.cofactorRow`
- Expansion of a row in coordinate vectors — `MathlibAnnex.Piola.sum_smul_basisRow`
- Determinant linearity over a finite sum in one row — `MathlibAnnex.Piola.det_updateRow_finset_sum`
- Cofactor expansion of a replaced row —  
`MathlibAnnex.Piola.det_updateRow_eq_sum_mul_cofactorRow`
- Perturbation of one output coordinate — `MathlibAnnex.Piola.singleOutputPerturb`
- Scalar times a cofactor field — `MathlibAnnex.Piola.componentFlux`
- Flux support contained in scalar support — `MathlibAnnex.Piola.support_componentFlux_subset`
- Compact support passes to the flux — `MathlibAnnex.Piola.hasCompactSupport_componentFlux`

## Lean source signature (exact)

```
theorem divergence_componentFlux_eq_det_sub
  {n : ℕ} {g : (Fin n → ℝ) → (Fin n → ℝ)} {ϕ : (Fin n → ℝ) → ℝ}
  (hg : ContDiff ℝ 2 g) (hϕ : ContDiff ℝ 1 ϕ)
  (i : Fin n) (x : (Fin n → ℝ)) :
  ∑ j, fderiv ℝ (componentFlux g ϕ i) x (Pi.single j 1) j =
```

```
LinearMap.det ((fderiv ℝ (singleOutputPerturb g ϕ i) x).toLinearMap) -
  LinearMap.det ((fderiv ℝ g x).toLinearMap)
```

| In the source                                                                                                                  | Mathematical meaning                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>{g : (Fin n → ℝ) → (Fin n → ℝ)} {ϕ : (Fin n → ℝ) → ℝ}</code>                                                             | The vector map $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and scalar function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ .                                                                              |
| <code>(hg : ContDiff ℝ 2 g) (hϕ : ContDiff ℝ 1 ϕ) (i : Fin n) (x : (Fin n → ℝ))</code>                                         | The assumptions are $g \in C^2$ , $\varphi \in C^1$ , with fixed output row $i$ and arbitrary point $x$ . No support condition is imposed.                                                                    |
| <code>componentFlux g ϕ i</code>                                                                                               | The vector field $V(x) = \varphi(x)C_i(Dg(x))$ , whose scalar components are $V_j(x) = \varphi(x)C_i(Dg(x))_j$ .                                                                                              |
| <code>∑ j, fderiv ℝ (componentFlux g ϕ i) x (Pi.single j 1)</code><br><code>j</code>                                           | The divergence $\sum_j (DV(x)e_j)_j = \operatorname{div} V(x)$ . The derivative acts on the whole vector field before its $j$ th component is taken.                                                          |
| In the source                                                                                                                  | Mathematical meaning                                                                                                                                                                                          |
| <code>singleOutputPerturb g ϕ i</code>                                                                                         | The perturbed vector map $g^{(i)}(x) = g(x) + \varphi(x)e_i$ , defined in the cited declaration.                                                                                                              |
| <code>LinearMap.det ((fderiv ℝ (singleOutputPerturb g ϕ i) x).toLinearMap) - LinearMap.det ((fderiv ℝ g x).toLinearMap)</code> | The determinant increment $\det Dg^{(i)}(x) - \det Dg(x)$ . Forgetting the continuity certificate of a derivative does not change its linear map. This is the complete right side of the divergence identity. |

#### ▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators Topology

namespace MathlibAnnex
namespace Piola
```

#### ▼ Exact content identity

Declaration: `MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`  
Accepted content SHA-256: `3cc9f8af908a62b6ffc37136c15eb3dfc455e4e7c035797703b06f063709005d`  
Accepted source guide SHA-256: `3395ad2e21acce2f08cab3a89d044343b4013e2c6fa5cf7128e7da416be2afdb`  
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`